

HW CLASS 6 (R FUNCTIONS)

Q6. Generalized protein-drug interaction function

```
# analyze_pdb_bfactors()
# Input:
#   pdb_id = a Protein Data Bank ID as a character string, for example "4AKE"
#
# What this function does / how to use it:
#   This function reads a PDB structure, keeps only chain A alpha-carbon atoms,
#   extracts their B-factors, and plots the B-factor profile.
#   Example use: analyze_pdb_bfactors("4AKE")
#
# Output:
#   A B-factor plot for the specified protein and the numeric B-factor vector
#   returned invisibly.

analyze_pdb_bfactors <- function(pdb_id) {
  library(bio3d)

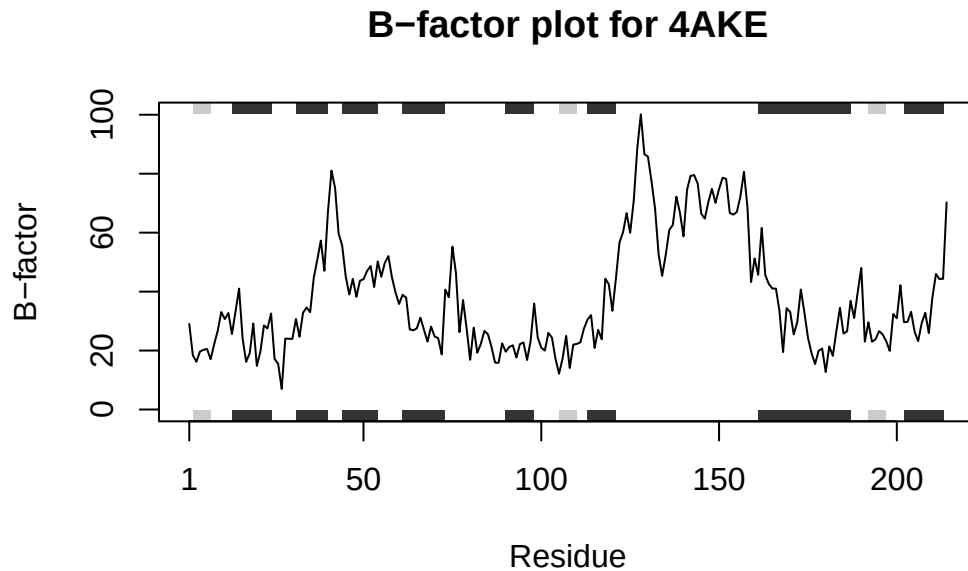
  pdb <- read.pdb(pdb_id)
  pdb_chainA <- trim.pdb(pdb, chain = "A", eley = "CA")
  b_factors <- pdb_chainA$atom$b

  plotb3(
    b_factors,
    sse = pdb_chainA,
    typ = "l",
    ylab = "B-factor",
    main = paste("B-factor plot for", pdb_id)
  )
}
```

```
invisible(b_factors)
}
```

```
analyze_pdb_bfactors("4AKE")
```

Note: Accessing on-line PDB file



This function takes a PDB ID as input, reads the corresponding protein structure, trims chain A to alpha-carbon atoms, extracts the B-factors, and plots the B-factor profile. The output is a B-factor plot for the selected protein, and the numeric B-factor vector is returned invisibly for later analysis.